

Stormwater Regulation Summary: Montgomery County, MD

Municipal Stormwater Management Technical Profile | Sempergreen USA

1. Regulatory Overview

- Primary driver: Chesapeake Bay TMDL compliance — Maryland has aggressive nutrient reduction targets.
- Environmental Site Design to Maximum Extent Practicable (ESD to MEP) is the core design philosophy.
- Maryland Department of Environment (MDE) Stormwater Design Manual + Montgomery County Code Chapter 19.
- Applies to ≥5,000 SF land disturbance; redevelopment triggers at ≥5,000 SF where existing imperviousness >40%.
- Montgomery County Building Energy Performance Standards (BEPS, adopted April 2024) apply to buildings ≥25,000 SF — energy efficiency requirements.

2. Typical Soil Conditions

- Montgomery County spans the Piedmont physiographic province — residual soils from metamorphic rock.
- Typical soils: Manor channery silt loam, Glenelg silt loam, Brinklow-Blocktown complex.
- Hydrologic Soil Group: Predominantly C, with some B soils in well-drained upland areas and D in valley bottoms.
- Piedmont clays limit infiltration in many areas — ESD to MEP acknowledges this by requiring design "to the maximum extent practicable."
- Implication: Many sites cannot fully achieve infiltration-based ESD — rooftop retention (green roofs, mineral wool) supplements ground-level BMPs.

3. Green Roof Requirements

- Green roofs are not explicitly mandatory but are a key ESD practice for meeting "Maximum Extent Practicable" standard.
- MDE Stormwater Design Manual lists green roofs as an approved ESD practice with specific sizing criteria.
- RainScapes Rebate Program: up to \$7,500 (residential) or \$20,000 (commercial/institutional) for green infrastructure, including green roofs.
- Montgomery County Green Bank offers Solar Loan program — supports bio-solar installations.
- IgCC (International Green Construction Code) adopted for commercial ≥5,000 SF — green roof can contribute to compliance.
- Purple Line corridor development is driving demand for rooftop BMPs on transit-oriented projects.

4. TSS Requirements

- Maryland requires 80% TSS removal for post-construction stormwater management.
- Green roofs are approved ESD practices that contribute to TSS removal for captured rainfall volume.
- Chesapeake Bay TMDL adds phosphorus and nitrogen reduction requirements — green roofs provide nutrient removal credit.
- MDE Stormwater Design Manual assigns pollutant removal efficiencies to each approved ESD practice.
- *Specific TSS and nutrient removal credits for green roofs should be confirmed in current MDE manual.*

5. Nature-Based Retention Requirements

- ESD to MEP: Replicate "Woods in Good Condition" (CN = 55) for the 1-year, 24-hour storm event.
- This is a high bar — the goal is to make the developed site perform hydrologically like an undisturbed forest.
- Retention sizing: Central/Eastern MD rainfall zone — 1.5" precipitation depth for WQv calculations.
- CN (Curve Number) method is the standard design approach — target CN 55 for ESD to MEP.
- Where achieving CN 55 is not practicable, demonstrate that ESD practices were applied to the maximum extent feasible.
- Mineral wool in green roof systems provides approximately double the retention capacity of soil alone — key for meeting the CN 55 target.

6. Allowable Outflow Rates

- Channel Protection (Cpv): 24-hour extended detention of the 1-year, 24-hour storm event.
- Overbank Flood Protection (Q10): 10-year storm peak controlled to pre-development rate.
- Extreme Flood (Q100): 100-year event safely conveyed; peak flow controlled to pre-development.
- Maryland does not specify a flat l/s/ha rate — control is relative to pre-development conditions.
- For reference, achieving CN 55 (woods in good condition) typically limits release to approximately 8-15 l/s/ha for the 10-year storm.
- *12-hour extended detention required in Use III/IV watersheds — stricter than the standard 24-hour requirement.*

Items in *italic* should be verified against current municipal codes. | Generated by Sempergreen USA Stormwater BMP Comparison Tool